

Dawei Xi

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Schmidt Science Fellow at University of California, Berkeley; incoming Assistant Professor at Harvard (2027)

Education and Experience

University of California, Berkeley, Berkeley, California, USA

- **Postdoctoral Fellow, Schmidt Science Fellow** July. 2025 – Present
- Membrane science, electrochemical interfaces, separation.

Harvard University, Cambridge, Massachusetts, USA

- **Postdoctoral Fellow** May. 2025 – July. 2025
- **Ph.D.** in Materials Science & Mechanical Engineering Sept. 2020 – May. 2025
- Carbon dioxide capture & utilization, flow battery, electrosynthesis, electrochemical systems.

University of Science and Technology of China (USTC), Hefei, Anhui, China

- **Master of Science** in Chemistry (Inorganic Chemistry) Sept. 2019 – Jun. 2021
- Electrocatalysis, CO₂ reduction, Joule heating.
- **Bachelor of Science (First Class Honored)** in Chemistry Sept. 2015 – Jun. 2019
- Nano synthesis, fabrication, characterization. **GPA: 3.96/4.30**, ranking **1st/156** in every academic year.

Selected Publications

- **Dawei Xi***, Panlin Zhao*, Manav Bansal, Olivia T. Vulpin, Shannon W. Boettcher#, Michael J. Aziz#. "Oxygen-Tolerant CO₂ Capture using Protected Redox-Driven Reverse Bias Bipolar Membrane Electrodialysis." *Carbon Future* (2026) (selected as the Cover).
- **Dawei Xi*#(corresponding)**, Yuheng Wu*, Yuli Li, Michael J. Aziz#. "Electrifying industrial hydrogen peroxide production via soft interfacial molecular mediation." *Nature Chemistry* (2025): 17.12: 1883-1890.
- **Dawei Xi*#(corresponding)**, Zheng Yang*, Michael Emmanuel, Michael J. Aziz#. "High efficiency acid-base generator for carbon management." *Energy & Environmental Science* (2025): 18 (8), 3839-3851.
- **Dawei Xi***, Abdulrahman M. Alfaraidi*, Jinxu Gao, Thomas Cochard, Luana CI Faria, Zheng Yang, Thomas Y. George et al. "Mild pH-decoupling aqueous flow battery with practical pH recovery." *Nature Energy* (2024): 9 (4), 479-490. (Highlighted by Nature Energy News & Views)
- **Dawei Xi**, Zheng Yang, Abdulrahman M. Alfaraidi, Yan Jing, Roy G. Gordon, Michael J. Aziz#. "Single-membrane pH-decoupling aqueous batteries using proton-coupled electrochemistry for pH recovery." *Energy Advances* (2024): 3 (8), 1911-1918.
- **Dawei Xi***, Jiayi Li*, Jingxiang Low*, Keke Mao, Ran Long, Jiawei Li, Zehui Dai Tianyi Shao, Yuan Zhong, Yu Li, Zibiao Li, Xian Jun Loh#, Li Song, Enyi Ye#, Yujie Xiong#. "Limiting the Uncoordinated N Species in M–Nx Single-Atom Catalysts toward Electrocatalytic CO₂ Reduction in Broad Voltage Range." *Advanced Materials* (2022): 34 (25), 2104090.
- Taobo Wang*, **Dawei Xi***, Michael J. Aziz#. "Electrochemical hydrogen storage under ambient conditions in aqueous-soluble organics." *ACS Applied Energy Materials* (2024): 7 (15), 6578-6584.
- Abdulrahman M. Alfaraidi*, **Dawei Xi***, Nina Ni, Thomas Y. George, Tatsuhiko Tsukamoto, Roy G. Gordon, Michael J. Aziz#, and Richard Y. Liu#. "An Extremely Stable and Soluble NH₂-Substituted Anthraquinone Electrolyte for Aqueous Redox Flow Batteries." *ACS Applied Energy Materials* (2023): 6 (24), 12259-12266.
- Xiaonong Wang*, Zhongliao Wang*, **Dawei Xi***, Jiayi Li, Xiaoxia Li, Xiujun Bai, Bin Wang, Jingxiang Low#, and Yujie Xiong#. "Tunable Impedance of Cobalt Loaded Carbon for Wide-Range Electromagnetic Wave Absorption." *Small* (2023): 20 (24), 2308970.
- Kiana Amini, Thomas Cochard, Yan Jing, Jordan D Sosa, **Dawei Xi**, Maia Alberts, Michael S Emanuel, Emily F Kerr, Roy G Gordon#, Michael J Aziz#. "In situ techniques for aqueous quinone-mediated electrochemical carbon capture and release." *Nature Chemical Engineering* (2024): 1 (12), 774-786.
- Wenqing Zhang, **Dawei Xi**, Yihong Chen, Aobo Chen, Yawen Jiang, Hengjie Liu, Zeyu Zhou et al. "Light-driven flow synthesis of acetic acid from methane with chemical looping." *Nature Communications* (2023): 14 (1), 3047.
- Zhen Zhang, **Dawei Xi**, Zhichu Ren, and Ju Li#. "A carbon-efficient bicarbonate electrolyzer." *Cell Reports Physical Science* (2023): 4 (11).
- Kiana Amini, Thomas Cochard, Yan Jing, Jordan D Sosa, **Dawei Xi**, Maia Alberts, Michael S Emanuel, Emily F Kerr, Roy G Gordon#, Michael J Aziz#. "In situ techniques for aqueous quinone-mediated electrochemical carbon capture and release." *Nature Chemical Engineering* (2024): 1 (12), 774-786.

- Shuai Pang*, Shijian Jin*, Fengcun Yang, Maia Alberts, Lu Li, **Dawei Xi**, Roy G. Gordon, Pan Wang#, Michael J. Aziz#, and Yunlong Ji#. "A phenazine-based high-capacity and high-stability electrochemical CO₂ capture cell with coupled electricity storage." *Nature Energy* (2023): 8 (10), 1126-1136.

Conference presentations

- Dawei Xi** (Invited), Electrifying nonaqueous chemistry with aqueous electrochemistry. International Symposium on Carbon Future & Catalysis, 2025.
- Dawei Xi** (Invited), Management of protons and electron transfer in aqueous electrochemistry for decoupled energy storage and multi-phase conversion. *eScience* Flow Battery Forum, 2025.
- Dawei Xi** et, al., Electrochemical acid-base generators for carbon capture. AIChE Annual Meeting, 2025.
- Dawei Xi** et, al., Interfacial proton-coupled electron transfer for electrifying nonaqueous chemistry. AIChE Annual Meeting, 2025.
- Dawei Xi** et, al., Aqueous Redox Active Molecular Materials Mediating Proton-Coupled-Electron-Transfer for Sustainable Energy Applications. MRS Spring Meeting, 2025.
- Dawei Xi** et, al., High efficiency acid-base generator for carbon management. MRS Fall Meeting, 2024.
- Dawei Xi** et, al., Electrifying industrial hydrogen peroxide production. Gordon Research Conference in Electrochemistry, 2024; Electrochemical Society Meeting Abstracts, 2024.
- Dawei Xi** et, al., Single-membrane pH-decoupling aqueous battery using proton-coupled electrochemistry for pH recovery. Electrochemical Society Meeting Abstracts, 2024.
- Dawei Xi** et, al., Mild pH-decoupling aqueous flow battery with practical pH recovery. Gordon Research Conference in Electrochemistry, 2024; Electrochemical Society Meeting Abstracts, 2023.
- Dawei Xi** et, al., Crossover management for practical high efficiency carbon dioxide reduction. Electrochemical Society Meeting Abstracts, 2024.

Patents

- Dawei Xi**, Michael J. Aziz, Yuheng Wu. System and method for production of hydrogen peroxide. Filed, Licensed by *Adiabatic Materials* (2025). Attorney Docket No: 51198-057WO2.
- Michael J. Aziz, Richard Y. Liu, **Dawei Xi**, Abdulrahman Alfaraidi. Multiphase low-energy CO₂ capture mediated by proton-coupled organic redox. Filed (2025). Attorney Docket No: 51198-064001.
- Dawei Xi**, Taobo Wang, Michael J. Aziz. Reversible hydrogen storage. Filed (2024). Attorney Docket No: 51198-043001.
- Dawei Xi**, Michael J. Aziz, Zheng Yang. pH-decoupled redox flow batteries. Filed (2024). Attorney Docket No: 51198-056001.
- Dawei Xi**, Michael Aziz. Asymmetric membranes for highly efficient CO₂ reduction. Filed (2024). Attorney Docket No: 51198-082001.
- Yujie Xiong, **Dawei Xi**, Jiayi Li, Ran Long. A method of synthesizing single atom catalysts using joule heating shock and its application in CO₂ reduction. ZL (2021) 1 0466113.0, 2024.

Selected Awards and Honors

Carbon Future Young Investigator Award, <i>Carbon Future Editorial Board</i>	May, 2025
MRS Graduate Student Award, <i>Materials Research Society</i>	Feb, 2025
eScience Distinguished Talk Award, <i>eScience Editorial Board</i>	Feb, 2025
Innovation Incentive Award, <i>Jiangsu Renewable Energy Society</i>	Dec, 2020
First-class Scholarship × 2, <i>Chinese Ministry of Education</i>	Sept. 2019, Sept. 2020
Selected in Outstanding Graduates Directory, <i>USTC</i>	Jun. 2019
Guo Moruo Scholarship (Highest Honor of USTC, Top 0.4% in USTC), <i>USTC</i>	Oct. 2018
Selected in National Scholarship Directory (Top 100 in China, ~Top 0.5% of National Scholarship Owners), <i>Chinese Ministry of Education, People's Daily</i>	Feb. 2018
National Scholarship × 2 (Top 1.5%), <i>Chinese Ministry of Education</i>	Sept. 2017, Sept. 2016
Chinese Chemistry Olympiad, Silver medal (Top 150 in China), <i>Chinese Chemical Society</i>	Dec. 2014